AI MULTILINGUAL TEXT SUMMARIZER

Project submitted for the partial fulfillment of the requirements for the course

CSE 455: Artificial Intelligence

Offered by the

Department Computer Science and Engineering

School of Engineering and Sciences

Submitted by

S. Jaswanth Sai Krishna- AP23110010961

N. Jayadeep -AP23110010934

P. Venkata Gopi -AP23110010994

Ch. Sathwik - AP23110010812

SRM University–AP

Neerukonda, Mangalagiri, Guntur

Andhra Pradesh – 522 240

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ABSTRACT

In the modern digital era, a massive amount of information is generated daily in the form of articles, reports, PDFs, and images. Reading and understanding all this content manually is time-consuming and inefficient. To address this issue, this project presents an AI-based multilingual text summarizer that can automatically extract and summarize information from various input formats.

The system is capable of processing plain text, PDF documents, and images. It uses Optical Character Recognition (OCR) to extract text from images and PDF parsing techniques to retrieve content from documents. After extraction, the system applies Natural Language Processing (NLP) techniques to generate concise summaries. Additionally, it supports multiple languages by integrating translation features, allowing users to view summaries in their preferred language.

The main objective of this project is to reduce reading effort, save time, and improve accessibility to information. The system also provides useful metrics such as word count and compression ratio, helping users understand how much the text has been reduced.

INTRODUCTION

With the rapid growth of digital content, people are exposed to a large amount of textual data every day. Whether it is academic materials, business reports, or online articles, users often find it difficult to go through lengthy documents and extract useful information quickly.

Traditional methods of reading and summarizing content require significant time and effort. This creates a need for automated systems that can perform summarization efficiently. The development of Artificial Intelligence, especially in Natural Language Processing, has made it possible to design such systems.

This project focuses on building an intelligent text summarizer that not only summarizes text but also supports multiple input formats such as PDFs and images. This makes the system more flexible and useful in real-world situations. For example, users can upload a scanned document or image, and the system will extract the text and summarize it automatically.

Another important feature of this project is multilingual support. Many users prefer to read content in their native language. By integrating translation features, the system allows users to convert summaries into languages such as Telugu, Hindi, and others.

Overall, this project aims to provide a user-friendly, efficient, and intelligent solution for handling large textual data.

PROBLEM STATEMENT

In today's fast-paced environment, users do not have enough time to read long documents completely. Important information is often hidden within large amounts of text, making it difficult to identify key points quickly.

Some of the major problems faced by users include:

- Difficulty in reading lengthy documents
- Time consumption in extracting important information
- Lack of tools that support multiple input formats
- Limited availability of multilingual summarization tools
- Inefficiency in handling large-scale text data

Existing summarization tools have certain limitations. Most of them only work with plain text and do not support PDF or image inputs. Some tools also fail to handle large documents due to memory or token limitations. Additionally, multilingual support is often limited or inaccurate.

Therefore, there is a need for a system that can:

- Accept different types of input (text, PDF, image)
- Extract text automatically from files
- Generate meaningful summaries
- Support multiple languages
- Handle large content efficiently without errors

This project aims to solve these problems by developing a complete and efficient summarization system.

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Functional Requirements

The system must perform the following functions:

- Allow users to input text manually through a text box
- Provide an option to upload PDF documents
- Allow users to upload image files (JPG, PNG, etc.)
- Extract text from PDFs using document processing techniques
- Extract text from images using OCR technology
- Clean and preprocess the extracted text
- Automatically detect the language of the input text
- Translate text when required
- Generate summaries based on user-selected length (short, medium, long)
- Display the summarized output clearly
- Show statistics such as original word count, summary word count, and compression ratio
- Provide an option to download the summary as a PDF

4.2 Non-Functional Requirements

The system should satisfy the following quality attributes:

- **Usability:** The interface should be simple and easy to use for all users
- **Performance:** The system should generate summaries within a reasonable time
- **Reliability:** It should handle errors such as invalid files gracefully
- **Scalability:** It should be able to handle large documents efficiently
- **Accuracy:** The summary should retain the important meaning of the original text

4.3 Software Requirements

The system is developed using the following technologies:

- Python programming language
- Streamlit for user interface
- Transformers library for summarization
- PyTorch for model execution

- OpenCV and PIL for image processing
- Tesseract OCR for text extraction from images
- PyMuPDF for PDF text extraction
- Googletrans for translation
- ReportLab for generating downloadable PDFs

4.4 Hardware Requirements

- A system with at least 8 GB RAM
- Processor capable of running Python applications
- Optional GPU for faster processing
- Internet connection for translation services

METHODOLOGY

5.1 System Architecture

The system is designed using a modular architecture consisting of multiple stages:

1. Input Stage
The user provides input either by typing text or uploading a file (PDF or image).

2. Extraction Stage
Depending on the input type:

- PDF files are processed to extract text from all pages
- Images are processed using OCR to detect and extract text

3. Preprocessing Stage
The extracted text is cleaned by:

- Removing unnecessary characters
- Normalizing spaces and formatting
- Ensuring the text is ready for processing

4. Processing Stage

- The system detects the language of the input
- If the text is not in English, it is translated into English
- The text is divided into smaller chunks to handle large inputs
- Each chunk is summarized using an AI model
- The summarized chunks are combined into a final summary

5. Output Stage

- The final summary is displayed to the user
- Metrics such as word count and compression ratio are shown
- The user can download the summary as a PDF

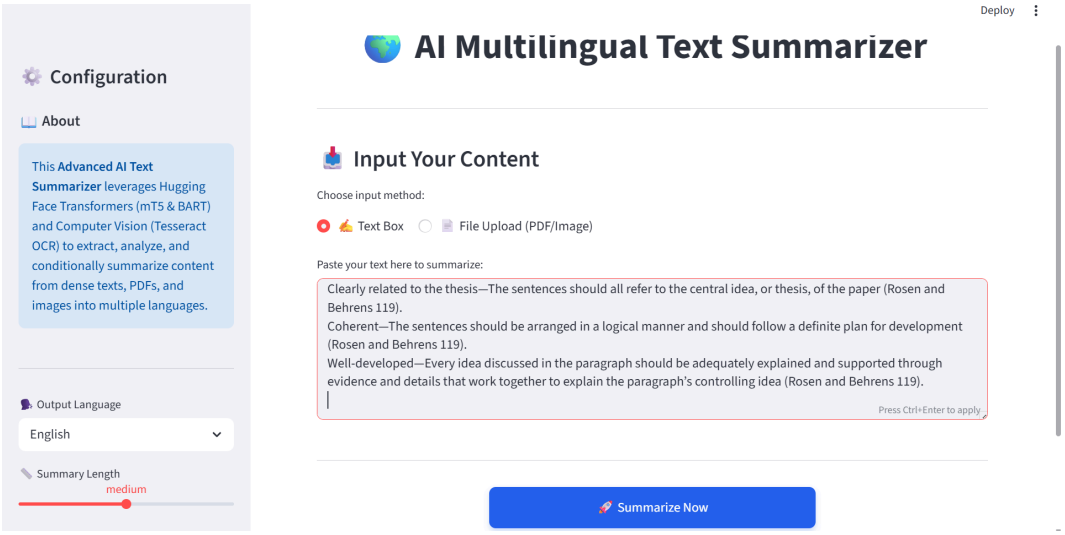
5.2 Algorithm Explanation

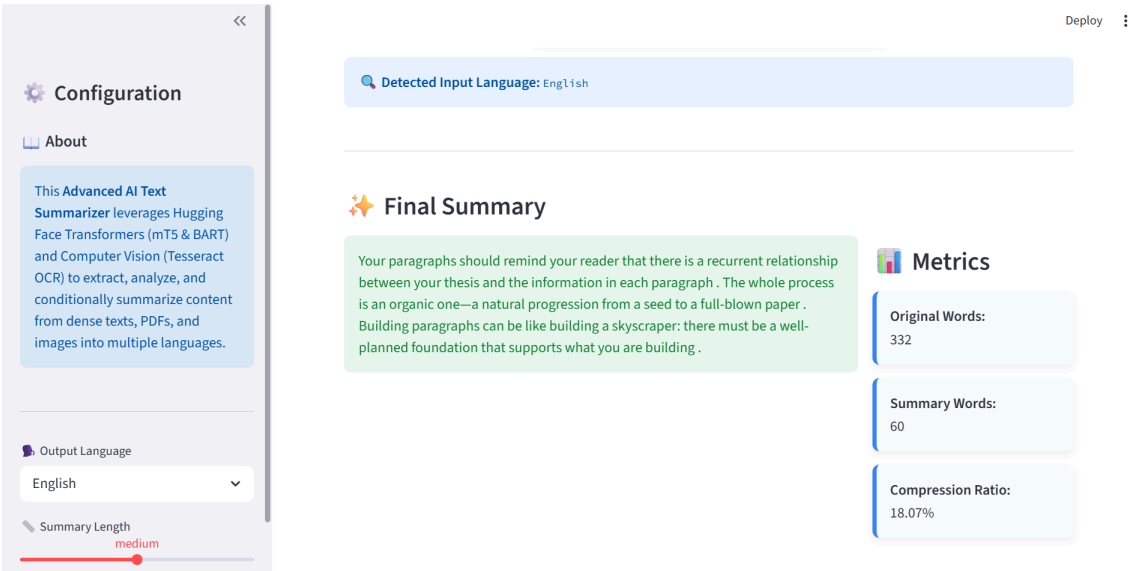
The algorithm used in this system follows a step-by-step approach:

1. **Input Collection**
   The system receives input from the user in the form of text, PDF, or image.

2. **Text Extraction**
   If the input is a file, the system extracts text using appropriate methods.
3. **Text Cleaning**
   The extracted text is cleaned to remove unwanted characters and formatting issues.
4. **Language Detection**
   The system identifies the language of the input text.
5. **Translation (if required)**
   If the input language is not English, it is translated into English for better processing.
6. **Chunking**
   The text is divided into smaller parts to avoid processing limitations.
7. **Summarization**
   Each chunk is processed using a transformer-based model to generate a summary.
8. **Combination**
   All summarized parts are combined to form the final summary.
9. **Final Translation**
   The summary is translated into the user's selected language if needed.
10. **Output Display**
    The final summary is shown along with metrics.

Test Cases


Deploy
Configuration
About
This Advanced AI Text Summarizer leverages Hugging Face Transformers (mT5 & BART) and Computer Vision (Tesseract OCR) to extract, analyze, and conditionally summarize content from dense texts, PDFs, and images into multiple languages.
Output Language
English
Summary Length
medium
AI Multilingual Text Summarizer
Input Your Content
Choose input method:
Text Box
File Upload (PDF/Image)
Paste your text here to summarize:
Clearly related to the thesis—The sentences should all refer to the central idea, or thesis, of the paper (Rosen and Behrens 119).
Coherent—The sentences should be arranged in a logical manner and should follow a definite plan for development (Rosen and Behrens 119).
Well-developed—Every idea discussed in the paragraph should be adequately explained and supported through evidence and details that work together to explain the paragraph's controlling idea (Rosen and Behrens 119).
Press Ctrl+Enter to apply
Summarize Now


Deploy
Configuration
About
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Output Language
English
Summary Length
medium
Detected Input Language: English
Final Summary
Your paragraphs should remind your reader that there is a recurrent relationship between your thesis and the information in each paragraph . The whole process is an organic one—a natural progression from a seed to a full-blown paper . Building paragraphs can be like building a skyscraper: there must be a well-planned foundation that supports what you are building .
Metrics
Original Words:
332
Summary Words:
60
Compression Ratio:
18.07%

Deploy

Configuration

About

This **Advanced AI Text Summarizer** leverages Hugging Face Transformers (mT5 & BART) and Computer Vision (Tesseract OCR) to extract, analyze, and conditionally summarize content from dense texts, PDFs, and images into multiple languages.

Output Language

Telugu

Summary Length

medium

Final Summary

మీ థీసిస్ మరియు ప్రతి పేరాలోని సమాచారానికి మధ్య పునరావృత సంబంధం ఉందని మీ పేరాగ్రాఫ్‌లు మీ పాఠకులకు గుర్తు చేయాలి.మొత్తం ప్రక్రియ సేంద్రియమైనది-విత్తనం నుండి పూర్తిస్థాయి కాగితం వరకు సహజమైన పురోగతి. పేరాగ్రాఫ్‌లను నిర్మించడం అనేది ఆకాశహర్మ్యాన్ని నిర్మించడం లాగా ఉంటుంది: మీరు నిర్మిస్తున్న దానికి మద్దతిచ్చే చక్కటి ప్రణాళికాబద్ధమైన పునాది ఉండాలి.

Metrics

Original Words:
332

Summary Words:
38

Compression Ratio:
11.45%

Download Summary as PDF

Deploy

Configuration

About

This **Advanced AI Text Summarizer** leverages Hugging Face Transformers (mT5 & BART) and Computer Vision (Tesseract OCR) to extract, analyze, and conditionally summarize content from dense texts, PDFs, and images into multiple languages.

Output Language

Telugu

Summary Length

medium

Text Box File Upload (PDF/Image)

Upload a document

Drag and drop file here
Limit 200MB per file • PDF, PNG, JPG, JPEG

Browse files

WHITE BOX TESTING.pdf 1.6MB

Text extracted successfully!

Preview Extracted Text

Summarize Now

Configuration

About

This **Advanced AI Text Summarizer** leverages Hugging Face Transformers (mT5 & BART) and Computer Vision (Tesseract OCR) to extract, analyze, and conditionally summarize content from dense texts, PDFs, and images into multiple languages.

Output Language

English

Summary Length

medium

Deploy

Final Summary

BASIS PATH & CONTROL STRUCTURE WHITE BOX TESTING MEMBERS P.VENKATA SRAVAN-AP23110010963 YASWANTH-AP 23110010812 INTRODUCTION White box testing focuses on testing the internal structure of a program . This technique is mainly used during unit testing to identify logical errors . Each node represents statements or decision points in the program. Edges represent the flow of control between nodes .

Metrics

Original Words:
378

Summary Words:
59

Compression Ratio:
15.61%

Download Summary as PDF

RESULTS

The system was tested using various types of inputs including short text, long documents, PDFs, and images. The results show that the system performs effectively in most cases.

- The OCR module successfully extracted text from clear images
- The PDF module handled multi-page documents efficiently
- The summarization model produced meaningful summaries
- The translation feature worked correctly for multiple languages
- The compression ratio helped users understand the reduction in content

For example:

- Original text: 1500 words
- Summary: 200–300 words
- Compression ratio: around 15–20%

This shows that the system significantly reduces content while preserving important information.